

Journal Choice

European Respiratory Journal (ERJ; European Respiratory Society)

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How to diagnose TB in migrants? A systematic review of reviews and decision tree analytical modelling exercise to evaluate properties for single and combined tuberculosis screening tests is a 2025 paper by Zenner et al., [published](#) in the European Respiratory Journal. This two-part study is a systematic review of reviews of diagnostic tests for active tuberculosis and a decision tree analysis of the pooled results. It concludes that “The significant test accuracy benefit of adding interferon- γ release assays to an active tuberculosis screening pathway will help inform clinicians and policy-makers on the most effective screening algorithms.” However, no cost effectiveness analysis was conducted. The paper explicitly calls for further economic analysis and exploration.

This work is a direct continuation and expansion of *How to diagnose TB in migrants?*, dealing with the same subject from a health economic perspective. It addresses the same problem and uses the data from the decision tree constructed in the second stage of the 2025 paper as input for a Markov cohort model to assess strategies’ cost-effectiveness.

Manuscript preparation

The thesis manuscript was structured as a draft journal paper: abstract, introduction, methods, results, discussion, and supplementary files. The CHEERS 2022 checklist for health economic evaluations was used to ensure high-quality reporting. The supplementary files include S1. Model Assumptions. Strengths and Limitations, S2. Screening Decision Tree/Initial Conditions for Markov Model, S3. Model Parameters, S4. Cost-Effectiveness Results, S5.

Sensitivity Analyses, and S6. Model Code. Vancouver style was used in citation formatting.

This manuscript will be used in preparing the final draft for the cost-effectiveness analysis paper. Several improvements listed in the Future Work section will be implemented prior to the final submission. The parametrisation and high-impact model assumptions will be reviewed by subject matter experts to assess their viability.